

Irish Math Olympiad 1988 — P12

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Prove that if n is a positive integer, then

$$\sum_{k=1}^n \cos^4\left(\frac{k\pi}{2n+1}\right) = \frac{6n-5}{16}$$

Solution

Where $z = e^{i\theta}$, we have $16 \cos^4 \theta = (z + z^{-1})^4 = (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$. So the sum becomes

$$\frac{6n}{16} + \frac{1}{16} \sum_{k=1}^n (2 \cos(2\phi_k) + 8 \cos(\phi_k))$$

where $\phi_k = 2k\pi/(2n+1)$.

We notice that $e^{i\phi_0}, e^{i\phi_1}, \dots, e^{i\phi_{2n}}$ form the $(2n+1)$ -th roots of unity. Since they solve the equation $x^{2n+1} - 1 = 0$ we deduce that they sum to 0 using Vieta's formulae. Also since $x^{2n+1} = e^{2m\pi ik} = e^{2\pi ik}$, we get that $x = e^{mi\phi_k} = e^{i\phi_k}$ for any positive integer m . Now,

$$\begin{aligned} \sum_{k=0}^{2n} \cos(m\phi_k) &= 0 \\ \sum_{k=1}^{2n} \cos(m\phi_k) &= -1 \end{aligned}$$

Note that $\cos(m\phi_{2n+1-k}) = \cos\left(2m\pi - \frac{2km\pi}{2n+1}\right) = \cos(2km\pi/(2n+1)) = \cos(m\phi_k)$ and so we can write

$$\sum_{k=1}^{2n} \cos(m\phi_k) = \sum_{k=1}^n [\cos(m\phi_k) + \cos(m\phi_{2n+1-k})] = \sum_{k=1}^n 2 \cos(m\phi_k) = -1$$

And so our original sum becomes

$$\frac{6n}{16} + \frac{1}{16}(-1 - 4) = \frac{6n-5}{16}$$

as desired. ■